Chapter 1

INTRODUCTION

Traditional composting is slow, requires manual effort and is not suitable for large-scale waste. To overcome these issues, we chose the Automatic Smart Compost Bin project. It aims to make waste management easier and more efficient by automating the composting process using sensors, IoT technology and smart control systems. This system not only reduces human effort but also converts organic waste into useful compost and liquid fertilizer.

1. Relevance of the Work

The proposed Automatic Compost Bin is highly relevant in today's world, where sustainable waste management and environmental protection are growing global priorities. With increasing urbanization and food waste generation, there is an urgent need for efficient household-level composting solutions. Traditional composting methods require continuous manual supervision and are time-consuming, often leading to unpleasant odour, pests, and improper decomposition.

By integrating IoT technology with sensors to monitor parameters like temperature, humidity, and gas emission, the proposed system automates and optimizes the composting process. The smart bin ensures proper microbial activity, reduces human effort, and enables users to convert organic waste into nutrient-rich compost efficiently. This solution contributes directly to smart city initiatives and promotes eco-friendly waste management practices.

This project is highly relevant in modern applications such as:

- Smart waste management and urban sustainability programs
- IoT-based home automation for environmental monitoring
- Rural and community-level composting initiatives

2. Issues and Challenges

- **Sensor Accuracy and Calibration:** Maintaining accurate readings from temperature, humidity, and gas sensors (e.g., DHT11, MQ135) is challenging due to variations in waste composition and environmental factors.
- **Odour and Gas Control:** Managing unpleasant odours and methane or ammonia gas buildup requires careful sensor-based ventilation control.
- **Power Management:** As the composting process takes several days, efficient power consumption and continuous operation of the microcontroller and sensors are crucial.
- **Moisture Balance:** Maintaining the ideal moisture content is critical for decomposition; under- or over-moist conditions can slow the process or cause foul smells.
- **System Integration:** Integrating multiple components (ESP32, motor drivers, sensors, and Wi-Fi module) with proper real-time control logic requires careful hardware and software synchronization.

3. Problem Statement and Objectives

3.1 Problem Statement:

Traditional composting methods are manual, time-consuming, and often inefficient due to lack of monitoring and control over critical parameters like temperature and moisture. Many urban households avoid composting due to odour issues and inconvenience. There is a need for an automated, low-cost composting system that ensures faster decomposition, odour control, and minimal user intervention through smart sensing and IoT connectivity.

3.2 Objectives:

- To convert organic waste into nutrient-rich compost.
- To promote sustainable waste management practices.
- To monitor and maintain ideal composting conditions.

4. Significant Original Contributions

- Developed an IoT-integrated composting system using ESP32, DHT11 and MQ135 sensors for real-time environmental monitoring.
- Implemented automatic control of aeration and moisture based on sensor feedback to ensure optimal decomposition.
- Designed a user-friendly mobile dashboard for real-time updates and compost status tracking.
- Designed a low-cost, portable prototype suitable for both household and community waste management.
- Contributed to sustainability goals by reducing landfill waste and promoting organic compost production

5. Potential Publication Areas:

This project can be published or presented in domains related to:

- **IoT-Based Smart Waste Management Systems** – focusing on automation and environmental monitoring using sensors and microcontrollers.
- **Sustainable and Green Technologies** – highlighting eco-friendly solutions for household and community-level composting.
- **Smart Agriculture and Soil Health Improvement** – showcasing how compost optimization contributes to soil enrichment and sustainable farming.
- **Renewable Energy and Environmental Engineering** – demonstrating efficient waste-to-resource conversion techniques.

Chapter 2

LITERATURE SURVEY

The increasing generation of kitchen and garden waste poses environmental and management challenges that demand sustainable solutions. Composting, when integrated with automation and IoT technology, provides an efficient and eco-friendly method for waste management. This literature review synthesizes key studies that have contributed to the understanding, design, and advancement of smart composting systems, with a particular emphasis on automation, sensor integration, and compost quality assessment [1].

Singh et al. demonstrated that food and kitchen waste can be effectively converted into biogas and organic fertilizer through anaerobic digestion. Their study established the principle that waste materials possess inherent resource value, transforming the perception of waste from garbage to usable energy and nutrients. While their focus was dual — energy and fertilizer production — it inspired later works (including the present project) that focus more narrowly on compost quality and automation rather than biogas generation [2].

Bhoir et al. introduced an IoT-based compost system capable of monitoring key composting parameters such as temperature, moisture, and aeration. This study provided foundational insights into how sensor data can enhance composting efficiency and uniformity. However, their system was primarily passive-focused on data collection and monitoring. The current project extends their work by automating composting actions such as shredding, mixing, and aeration, thereby closing the loop between sensing and actuation [3].

Saiyara et al. analysed the performance of organic waste composting and its agricultural applications. Their research confirmed that composting improves soil fertility and decreases dependency on chemical fertilizers. While the study was agricultural in nature, its findings validated the environmental and agronomic importance of composting as a sustainable waste management technique, providing the ecological justification for the development of advanced compost systems [4].

Badotra et al. developed an IoT-based waste segregation system that could automatically distinguish between wet, dry, and recyclable materials. This study emphasized the importance of pre-segregation to improve downstream composting efficiency [5].

Similarly, Goel et al. proposed an IoT-based garbage segregator with fill-level detection and notification capabilities, improving waste management logistics. Together, these works provided crucial insights into waste classification and monitoring, influencing the integration of intelligent pre-segregation into compost automation systems [6].

Li et al. examined the role of different bulking agents in aerobic composting, concluding that fallen leaves outperform tree prunings in processing kitchen waste due to better porosity and aeration. This study informed the use of garden waste as a complementary component in composting, highlighting the potential of resource synergy between kitchen and yard waste [7].

Adhikary et al presented an NPK sensor-based approach for real-time soil nutrient monitoring to advance precision agriculture. While their focus was on soil rather than compost, their methodology inspired the incorporation of similar sensors into composting systems to evaluate compost nutrient quality during processing [8].

Recent works have focused on the miniaturization and integration of IoT technologies in composting. Raghunath et aldeveloped a sensor-integrated smart composting bin for household use, showing how real-time monitoring minimizes odour and improves compost efficiency [9].

Likewise, Rajamuneeswaran et al. introduced the Smart Compost Guardian, a real-time alert system for compost management. Although both systems are primarily designed for small-scale or household applications, they underscore the scalability potential of IoT solutions for larger, automated composting operations [10].

Chapter 3

PROPOSED METHODOLOGY

The proposed system employs an ESP32 microcontroller as the main processing unit to monitor and control key parameters within the compost bin. A DHT11 sensor continuously measures the temperature and humidity levels to ensure optimal conditions for microbial activity. The collected data are processed by the ESP32, which determines whether the compost environment requires adjustments such as increased aeration or moisture.

A moisture sensor is integrated to detect the water content in the compost mixture. When the moisture level falls below the desired threshold, the system activates a mini water pump or spray mechanism to maintain ideal conditions for decomposition. Similarly, a motorized lid or fan can be controlled to improve airflow when the temperature rises beyond optimal limits.

All sensor readings are transmitted to a Blynk/IoT dashboard via Wi-Fi, enabling users to monitor the composting process remotely in real time. The interface also provides visual indicators and alert notifications for abnormal conditions. The system operates on a solar-powered battery setup, ensuring energy efficiency and uninterrupted functionality for sustainable waste management.

Chapter 4

RESULTS AND EVALUATION

Our project delivers a fully automated multi-chamber composting system that can monitor lid position, detect material levels, manage timed motor rotations and control servos for transferring compost between stages. It responds intelligently to environmental conditions inside the chambers, ensures mixing and movement happen only when required and maintains a smooth flow of compost through each phase of the process. The system integrates motors, servos and sensing mechanisms to maintain compost quality while reducing the need for manual intervention.

The figure 4.1 shows the design and main components of the automatic compost bin. It includes sensors such as the DHT22 for temperature and humidity, and a moisture sensor to monitor compost conditions. A shredder and fan help in aeration, while the mixing auger ensures proper blending of waste materials. The drainage system removes excess liquid, and the compost collector stores the final compost.

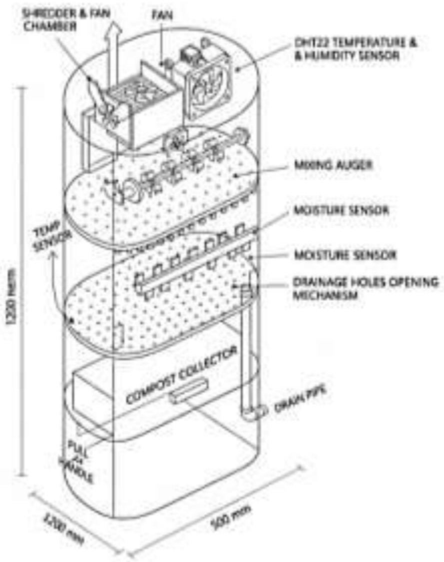

Figure 4.1: CAD model of automatic smart compost bin

The figure 4.2 shows the real-time monitoring screen of the IoT-based automatic compost bin. The interface displays important parameters such as temperature, soil moisture, and compost level. These readings help users track the composting process and ensure optimal conditions for decomposition. The data is collected through sensors and displayed on the mobile dashboard, allowing easy and efficient monitoring of the compost bin.

Figure 4.2: IOT based Compost Bin Status Screen

The figure 4.3 illustrates the sequential process involved in the automated compost bin system, beginning from waste collection and passing through the waste shredding unit, decomposition chamber, and sensor monitoring system. The data from the sensors are sent to the cloud dashboard for remote monitoring and alert generation. The automatic control system then manages the separation and collection of solid compost and liquid fertilizer, ensuring efficient composting and resource recovery.

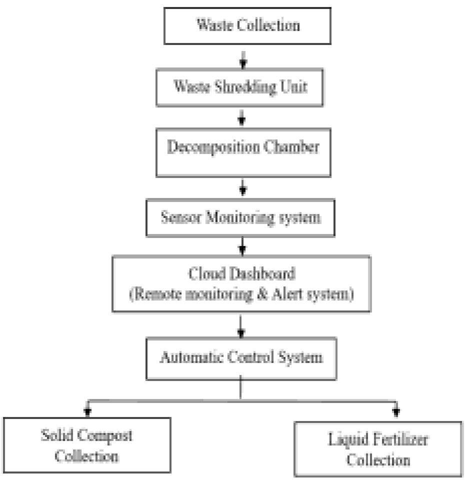

Figure 4.3: Flowchart Representation of Automated Compost Bin

The system works in four stages. In the **1st chamber**, Ultrasonic-1 checks the lid: if the distance is below 5 cm, Motor-1 runs for 5 minutes in both directions, and it stops whenever the distance goes above 5 cm, restarting only when the lid closes again. After each 5-minute cycle the lid must open once before the next cycle is allowed. Ultrasonic-2 checks if the chamber is full; when its distance goes below 5 cm the servo opens 45° to drop compost to chamber-2, waits 1 minute, then closes, and a Blynk alert warns the user to stop feeding. In the **2nd chamber**, Motor-2 mixes only when moisture is high, running every 2 hours for 10 minutes in both directions. If the temperature falls below 25 °C, Motor-2 stops and Servo-2 opens 45° to transfer compost to the 3rd chamber, then closes after 2 minutes. Moisture and temperature values are shown in Blynk. In the **3rd chamber**, Ultrasonic-3 triggers a Blynk notification when the compost level reaches 5 cm. In the **4th chamber**, an analog float sensor monitors liquid level and sends a Blynk alert when the chamber is full.

Figure 4.4: Overall Model of Automated Compost Bin

Chapter 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

The proposed automated smart compost pit system effectively demonstrates the use of IoT and embedded technology to optimize the composting process through real-time monitoring and control. By utilizing sensors to continuously track temperature and humidity, the system maintains optimal conditions for efficient organic waste decomposition. The ESP32 microcontroller automates aeration and moisture control, reducing manual labour and improving compost quality. The prototype is cost-effective, easy to operate and environmentally friendly, making it suitable for both domestic and small-scale commercial composting applications. Overall, this project contributes to sustainable waste management and promotes eco-friendly organic fertilizer production.

5.2 Future Works

- Adding more sensors to detect gases like methane and carbon dioxide for better monitoring of compost maturity and safety.
- Developing a mobile application to provide real-time monitoring, alerts and user-friendly data visualization.
- Improving the energy system by optimizing solar power and battery usage for longer, uninterrupted operation.
- Incorporating artificial intelligence for predictive analysis and automated control to optimize the composting process.
- Expanding the system to manage multiple compost bins simultaneously for larger-scale waste management.

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